

Toka Priyanka

tokapriyanka@gmail.com

AI/ML intern

DAY 3

Task 2

Body Measurement Logic With Human Pose Detection

Introduction:

Body Measurement Logic refers to the process of using Artificial Intelligence models to estimate different body measurements based on landmarks on the human body.

By examining the coordinates of the human body landmarks identified by AI models, it is possible to measure body dimensions such as shoulder width, height, and waist size.

Body measurement estimation is currently popular in different areas such as: Virtual clothing try-on services, Fashion suggestion applications, Fitness tracking applications, Health monitoring, Apparel sizing

Using frameworks such as MediaPipe, the AI algorithm can detect key points of the human body and then use them to estimate body measurements.

Measurement of Shoulder Width:

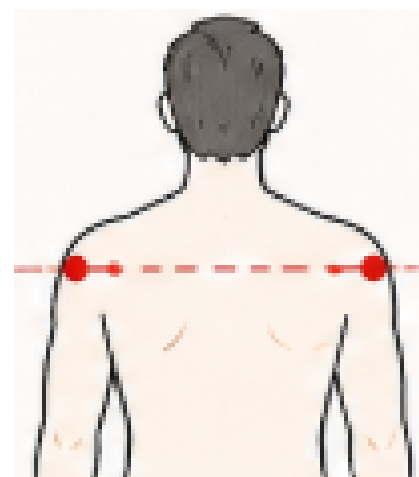
Description

The measurement of shoulder width is achieved by measuring the distance between the left shoulder landmark and the right shoulder landmark.

After detecting both shoulder landmarks, the AI algorithm can measure the distance between the two points.

Method of Operation

1. The pose estimation model detects the left shoulder landmark.
2. The AI algorithm detects the right shoulder landmark.
3. The distance between the two points is measured.
4. The calculated distance is used as an approximation of shoulder width.



Benefits of Shoulder Width Estimation

- Posture Monitoring

- **Personalized Fashion Systems**

It allows AI systems to estimate upper body proportions correctly.

Uses of shoulder width estimation include: Clothing recommendations, Virtual fitting rooms, Fitness tracking

Height Estimation:

Description

Height estimation refers to the approximation of a human body height through the detection of certain landmarks on the body, which is made possible through pose estimation.

In this case, the distance between landmarks found on the upper body such as the head and the landmarks located at the bottom parts such as the feet is measured.

How It Works

1. The pose detection system detects the topmost body landmark such as the head or nose.
2. The system then detects the bottom body landmarks such as ankles or feet.
3. The vertical distance between these two points is measured.
4. Finally, the distance estimated is used to approximate the body height.



Issues with Height Estimation

Height estimation may be complicated by: Camera angle, Distance between the camera and the body, Body posture, Image scaling, Lighting condition

Some systems employ camera calibration and depth estimation to achieve higher levels of accuracy.

Applications of Height Estimation

Height estimation finds application in: Fitness applications, Health monitoring systems, Avatar creation, E-commerce fashion, Biometric analysis

Estimation of Waist:

Introduction

The process of waist estimation is carried out based on the detection of body landmarks related to the hips and the waist by the pose estimation algorithm.

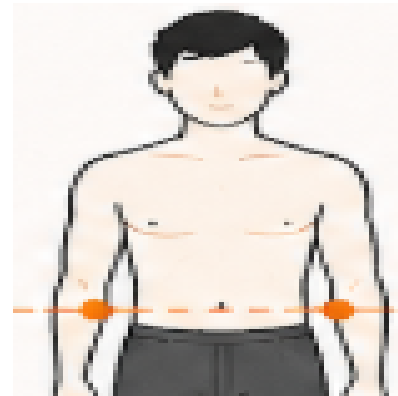
How it works

- 1.The AI system detects landmarks corresponding to the left and right hips.
- 2.The distance between the landmarks along the horizontal line is estimated.
- 3.The value is adjusted properly to measure the waist.

In some cases, more advanced artificial intelligence solutions include:

- Bodily segmentation
- Depth measurement
- 3D modeling of the body

to increase waist estimation accuracy.



Why Is Waist Estimation Important?

Waist estimation is essential in the following fields: Virtual try-on, Fashion recommendation systems, Health monitoring and personal training, Body shape analysis, Personalized clothing services

How Is AI Used in Body Measurements?

Artificial intelligence technologies are used for automatic estimation of body measurements without manual calculations. By applying pose estimation algorithms, AI systems are able to analyze body structures and estimate measurements instantly.

This makes it possible to increase: Convenience of the user, Experience in online shopping, Accuracy of recommendations

AI body measurement technology is gaining increasing significance in the modern fashion industry and health care.

Benefits of Body Measurement via Artificial Intelligence

- Nontactile measurement estimation
- Real-time body measurements
- Efficient processing time
- Better user interface design
- Suitable for virtual clothing fitting purposes
- Minimizes manual measurement errors

Conclusion:

The concept of Body Measurement Logic is crucial in AI-based human body analysis tools. With the help of body landmarks found via pose detection techniques, efficient measurement estimation of various parameters like shoulder width, body height, waist circumference, etc., can be performed. Such measurements have several practical applications in virtual clothing fitting solutions, fitness tracking devices, healthcare diagnosis systems, and fashion recommendation engines.